

EXPECTATION PROBLEMS

1. A coin is tossed until a head appears. Find the expected number of tosses.

The probability distribution of X is

outcomes	H	TH	TTH	TTTH.....
No. of tosses	1	2	3	4.....
Probability	$\frac{1}{2}$	$(\frac{1}{2})^2$	$(\frac{1}{2})^3$	$(\frac{1}{2})^4$

$$\begin{aligned}
 E(X) &= \sum_x xP(x) = 1(1/2) + 2(1/2)^2 + 3(1/2)^3 + 4(1/2)^4 + \dots \\
 &= 1/2(1 + 2(1/2) + 3(1/2)^2 + \dots) \\
 &= 1/2 \bullet (1 - 1/2)^{-2} \quad (1-x)^{-2} = 1 + 2x + 3x^2 + \dots \\
 &= 1/2 \bullet (1/2)^{-2} \\
 &= 1/2 \bullet (4) = 2
 \end{aligned}$$

2. From a box containing 2 red balls and 6 green balls. 2 balls are selected at random. Find the expected number of red balls that may be selected.

Number of red balls:

X	0(2G)	1(1G1R)	2(2R)
P(x)	0	1/2	1

$$\begin{aligned}
 &= \frac{6C_2}{8C_2} \quad \frac{2C_1 \times 6C_1}{8C_2} \quad \frac{2C_2}{8C_2} \\
 &= \frac{15}{28} \quad \frac{12}{8} \quad \frac{1}{28}
 \end{aligned}$$

$$E(X) = \sum xP(x) = 0 \times \frac{15}{28} + 1 \times \frac{12}{28} + 2 \times \frac{1}{28} = \frac{1}{2} \text{ Dr.K.Kavitha(SAS)}$$

3. Let X be a R.V with $E(X)=10$ and $V(X)=25$. Find the positive values of 'a' and 'b' such that $Y=aX-b$ has expectation zero and variance 1.

$$\text{Given } E(X)=10, V(x)=25$$

$$\text{Now, } E(Y)=E(aX-b)= aE(X)-b=0$$

$$a(10)-b=0\text{-----}(1)$$

$$\text{Var}(Y)=\text{Var}(aX-b)$$

$$=a^2\text{var}(x)$$

$$a^2(25)=1\text{-----}(2)$$

Solving eqns (1) and (2) , we get

$$a=1/5; b=2.$$

CO-VARIANCE (continuous case)

Problems:

1. Two random variables X and Y have the joint pdf.

$$f(x,y)=\begin{cases} \frac{xy}{96}, & 0 < x < 4, 1 < y < 5 \\ 0, & \text{otherwise} \end{cases} \quad \text{Find Cov}(X,Y).$$

Marginal density of X .

$$f(x)=\int_1^5 f(x,y)dy=\int_1^5 \frac{xy}{96}dy = x/8, 0 < x < 4$$

Marginal density of Y :

$$f(y)=\int_0^4 f(x,y)dx=\int_0^4 \frac{xy}{96}dx = y/12, 1 < y < 5 \quad \text{Dr.K.Kavitha(SAS)}$$

$$E(X) = \int_0^4 xf(x)dx = \int_0^4 x \frac{x}{8} dx = \frac{1}{24}(64) = 8/3$$

$$E(Y) = \int_1^5 yf(y)dy = \int_1^5 y \cdot \frac{y}{12} dy = 31/9$$

$$E(XY) = \int_0^4 \int_1^5 xyf(x,y)dydx = \int_0^4 \int_1^5 xy \frac{xy}{96} dydx$$

$$= \frac{1}{96} \int_0^4 x^2 (y^3/3)_1^5 dx = 248/27$$

$$\text{cov}(X,Y) = E(XY) - E(X)E(Y) \\ = \frac{248}{27} - \frac{8}{3} \cdot \frac{31}{9} = 0$$

CO-VARIANCE (Discrete case)

1. The Joint pmf of X and Y is

X \ Y	-1	1
0	1/8	3/8
1	2/8	2/8

Find the covariance of X and Y.

Marginal Pmf of X:

X	0	1
P(X)	4/8	4/8

Marginal Pmf of Y

Y	-1	1
P(Y)	3/8	5/8

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$$E(X) = \sum_x xP(x) = 4/8$$

$$E(Y) = \sum_y yP(y) = 2/8$$

$$E(XY) = \sum_x \sum_y xyP(x, y) = 0(1/8) + 0(3/8) + (-1)(2/8) + 1(2/8) = 0$$

$$COV(X, Y) = E(XY) - E(X)E(Y) = -1/8$$

Correlation

Correlation is a statistical tool which measures the degree of relationship between any two variable characteristics.

Karl Pearson's coefficient of correlation:

It is denoted as 'r' and is computed as,

$$r(x, y) = \frac{COV(X, Y)}{\sqrt{V(X)}\sqrt{V(Y)}} = \frac{E(XY) - E(X)E(Y)}{\sqrt{V(X)}\sqrt{V(Y)}}$$

Problem(Continuous case)

1. Two random variables X and Y have joint pdf

$$f(x, y) = \begin{cases} 2 - x - y, & 0 < x < 1, \quad 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Show that the correlation $r(x, y) = -1/11$

Marginal density of X.

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$$f(x) = \int_0^1 f(x, y) dy = \int_0^1 (2 - x - y) dy = 3/2 - x, 0 < x < 1$$

Marginal density of Y:

$$f(y) = \int_0^1 f(x, y) dx = \int_0^1 (2 - x - y) dx = \frac{3}{2} - y, 0 < y < 1$$

$$E(X) = \int_0^1 xf(x) dx = \int_0^1 x \left(\frac{3}{2} - x \right) dx = 5/12$$

$$E(Y) = \int_0^1 yf(y) dy = \int_0^1 y \left(\frac{3}{2} - y \right) dy = 5/12$$

$$E(X^2) = \int_0^1 x^2 f(x) dx = \int_0^1 x^2 \left(\frac{3}{2} - x \right) dx = 1/4$$

$$E(Y^2) = \int_0^1 y^2 f(y) dy = 1/4$$

$$E(XY) = \int_0^1 \int_0^1 xyf(x, y) dy dx = \int_0^1 \int_0^1 xy(2 - x - y) dy dx = 1/6$$

$$\text{cov}(X, Y) = E(XY) - E(X)E(Y) = -1/144$$

$$V(X) = E(X^2) - \{E(X)\}^2 = \frac{11}{144}$$

$$V(Y) = E(Y^2) - \{E(Y)\}^2 = \frac{11}{144}$$

$$r = \frac{\text{COV}(x, y)}{\sqrt{V(X)}\sqrt{V(Y)}} = \frac{-1}{11}$$

Problem(Discrete case)

1. The Joint pmf of X and Y is

Y \ X	-1	1
0	1/8	3/8
1	2/8	2/8

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Find the correlation correlation of X and Y.

Marginal Pmf of X:

X	0	1
P(X)	4/8	4/8

Marginal Pmf of Y

Y	-1	1
P(X)	3/8	5/8

$$E(X) = \sum_x xP(x) = 4/8$$

$$E(Y) = \sum_y yP(y) = 2/8$$

$$E(X^2) = \sum x^2 P(x) = 4/8$$

$$E(Y^2) = \sum y^2 P(y) = 1$$

$$V(X) = E(X^2) - \{E(X)\}^2 = 1/4$$

$$V(Y) = E(Y^2) - \{E(Y)\}^2 = 15/16$$

$$E(XY) = \sum_x \sum_y xyP(x, y) = 0(1/8) + 0(3/8) + (-1)(2/8) + 1(2/8) = 0$$

$$COV(X, Y) = E(XY) - E(X)E(Y) = -1/8$$

$$r = \frac{\text{cov}(X, Y)}{\sqrt{V(X)}\sqrt{V(Y)}} = -0.26$$

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Correlation: For frequency distribution the formula is

$$r = \frac{\frac{\sum xy}{n} - \bar{x} \bar{y}}{\sigma_x \sigma_y}, \text{ when the series is individual series}$$

For grouped data,2

$$r = \frac{\frac{\sum fxy}{\sum f} - \bar{x} \bar{y}}{\sigma_x \sigma_y} \text{ where } \sigma_x = \sqrt{\frac{1}{n} \sum x^2 - \bar{x}^2}; \quad \sigma_y = \sqrt{\frac{1}{n} \sum y^2 - \bar{y}^2}$$

Correlation coefficient (r) lies between -1 and 1.

Problem:

1. Calculate the correlation coefficient for the following heights(in inches) of fathers X and their sons Y.

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

X	Y	XY	X ²	Y ²
65	67	4355	4225	4489
66	68	4488	4356	4624
67	65	4355	4489	4225
67	68	4556	4489	4624
68	72	4896	4624	5184
69	72	4968	4761	5184
70	69	4830	4900	4761
72	71	5112	5184	5041
$\sum X = 544$	$\sum Y = 552$	$\sum XY = 37560$	$\sum X^2 = 37028$	$\sum Y^2 = 38132$

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$$\bar{X} = \frac{\sum X}{n} = \frac{544}{8} = 68$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{552}{8} = 69$$

$$\bar{X}\bar{Y} = 4962$$

$$\sigma_x = \sqrt{\frac{1}{n} \sum x^2 - \bar{x}^2} = \sqrt{\frac{37028}{8} - 4624} = 2.121$$

$$\sigma_y = \sqrt{\frac{1}{n} \sum y^2 - \bar{y}^2} = \sqrt{\frac{38132}{8} - 4761} = 2.345$$

$$r = \frac{\frac{\sum xy}{n} - \bar{x}\bar{y}}{\sigma_x \sigma_y} = \frac{\frac{37560}{8} - 4962}{2.121 \times 2.345} = 0.6030$$

2. Find the correlation coefficient between X and Y.

X	1	2	3	4	5
Y	2	5	3	8	7

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